Leveraging Causal Inference Methods for Detecting Distribution Shift

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Motivation

- In industry, huge amounts of data are generated incrementally where it comes in batches - existing batch of data $\mathbf{D}_{exist} = (\mathbf{X}_{exist}, Y_{exist})$ and new batch of data $\mathbf{D}_{new} = (\mathbf{X}_{new}, Y_{new})$. Types of shifts can vary:
  - Covariate Shift: $P(\mathbf{X}_{exist}) \neq P(\mathbf{X}_{new})$ where $P(Y_{exist}|\mathbf{X}_{exist}) = P(Y_{new}|\mathbf{X}_{new})$, only covariates distribution change where the conditional dependence remain the same.
  - Concept Drift: $P(Y_{exist}|\mathbf{X}_{exist}) \neq P(Y_{new}|\mathbf{X}_{new})$ where $P(\mathbf{X}_{exist}) = P(\mathbf{X}_{new})$, only conditional dependence change where the covariate distribution remain the same.
  - Distribution Shift: At least one of $P(Y|\mathbf{X})$ and $P(\mathbf{X})$ changes or both changes across existing batch of data and new batch of data.
- In practice, both types of shifts frequently arise simultaneously. **Existing methods** typically address only a specific type of shift. Our goal is to develop a hypothesis testing procedure that remains effective across a broad range of scenarios.

Statistical Methodology

- We propose a *class* of hypothesis tests for detecting the *distribution shift* $H_0 : P(\mathbf{X}_{new}) = P(\mathbf{X}_{exist}) \& P(Y_{new}|\mathbf{X}_{new}) = P(Y_{exist}|\mathbf{X}_{exist})$ $v.s.$ $H_a : P(\mathbf{X}_{new}) \neq P(\mathbf{X}_{exist})$ or $P(Y_{new}|\mathbf{X}_{new}) \neq P(Y_{exist}|\mathbf{X}_{exist})$ via leveraging causal inference methods.
- **We treat the existing batch of data $D_{exist} = (\mathbf{X}_{exist}, Y_{exist})$ as control group with batch assignment $T = 0$ and the new batch of data $D_{new} = (\mathbf{X}_{new}, Y_{new})$ as treatment group with treatment assignment $T = 1$.** Then the variable importance will represent the relative degree of change of $Y|\mathbf{X}$. *Figure 1* demonstrates the efficiency for recovering the ranking of the shifted features in both covariate shift and concept drift.

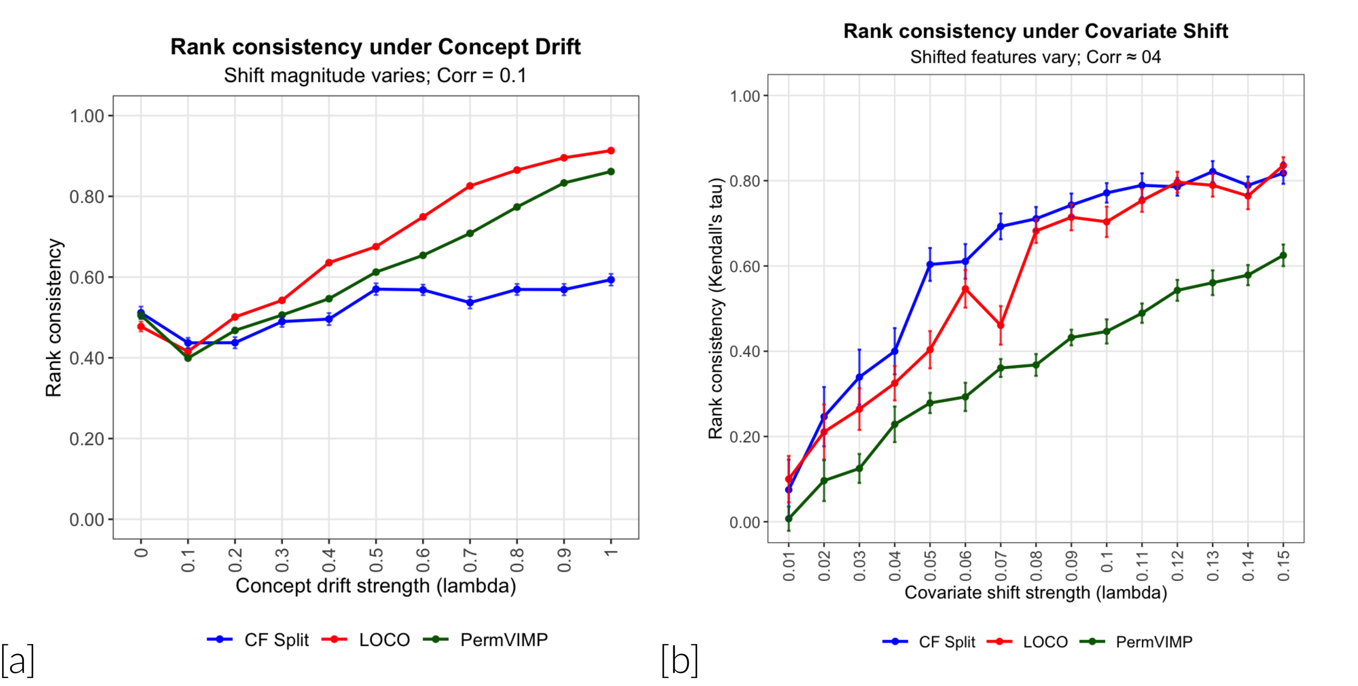

Figure 1. Illustration of the validity of the Variable Importance - the kendall τ correlation for ranking consistency of recovery of the degree of contribution to distribution shift.

- Existing batch of Data $D_{exist} : Y = X_1 + ... + X_5 + X_6 + ... + X_{10} + \epsilon$ and New batch of Data $D_{new} : Y = X_1 + 2 * X_2 - 3 * X_3 + 4 * X_4 - 5 * X_5 + 6 * X_6 - 7 * X_7 + 8 * X_8 + \epsilon$.
- Causal Forest(Athey, Tibshirani, and Wager 2019) is adapted, Confidence Interval of Feature Importance is displayed in Figure 1. The VIMP(Variable Importance) is approximately proportional to the degree of change of data-generating process $Y|\mathbf{X}$.
- The methodology can be summarized as a general-purposed three-step procedure, the details can be found in **Algorithm 1**.
  - First fit a causal learner and record the (original)variable importance.
  - Permute the Batch Assignment between the two batches of data for **B** times, retrain the causal learner and record the permuted variable importance.
  - Perform the hypothesis test based on whether there exist some features whose original variable importance is larger than all of the permuted variable importance.

The decision threshold of the test is decided by upper quantile of the variable importance.

Algorithm 1 Testing Covariate Shift via Permuting Treatment Assignment - **CFPerm**

BEGIN Start with existing batch of data $\mathcal{D}_{exist} = \{(\mathbf{X}_1, Y_1), ..., (\mathbf{X}_{n_{exist}}, Y_{n_{exist}})\}$ where $(\mathbf{X}_i, Y_i)$ and new batch of data $\mathcal{D}_{new} = \{(\mathbf{X}_{n_{exist}+1}, Y_{n_{exist}+1}), ..., (\mathbf{X}_{n_{exist}+n_{new}}, Y_{n_{exist}+n_{new}})\}$ where $(\mathbf{X}_i, Y_i)$, with each $\mathbf{X}_j = (x_1, ..., x_p)$. Assign (control) treatment $T = 0$ to existing batch of data and $T = 1$ (treatment) to new batch of data. Define hypotheses $H_0 : P(\mathbf{X}_{new}, Y_{new}) = P(\mathbf{X}_{exist}, Y_{exist})$ vs. $H_a : P(\mathbf{X}_{new}, Y_{new}) \neq P(\mathbf{X}_{exist}, Y_{exist})$.

S1 Fit the causal forest on $(\mathbf{X}, Y, T)$ and record the variable importance (VIMP) for each feature $v_1, ..., v_p$ (VIMP can be user-specified as cover based or variance based)

S2 Permute the treatment assignment T randomly across the data a total of B times (by default $B = 500$). Each time, fit the causal forest on the resulting permuted batches and calculate the variable importance, recorded as $v_{b,1}, ..., v_{b,p}$ for $b = 1, .., B$.

S3 Construct an interval for the variable importance of each of the feature with the lower bound as the smallest value and the higher bound as the largest value across the permutations $I_i = (v_{i,min}, v_{i,max}), i = 1, 2, ..., p$

S4-1 **CFPerm0**: The maximal value of the permuted confidence interval of variable importance for each of the feature is recorded as $I_{max} = \{v_{max,1}, ..., v_{max,p}\}$. Reject H_0 if there exists a feature whose original variable importance is larger than the maximal of the upper bound among all of the features $max(I_{max})$ served as the decision threshold; otherwise, do not reject.

S4-2 **CFPerm1**: The 99% upper quantile for the permuted confidence interval of variable importance in each of the feature is recorded: $I_{99\%} = \{v_{1,99\%}, ..., v_{p,99\%}\}$ Reject H_0 based on whether there exists a feature whose original variable importance is larger than the 95% upper quantile $q_{0.95}(I_{99\%})$ of the recorded feature-wise 99% quantile across the features served as the decision threshold. Otherwise the null hypothesis would not be rejected.

END

Figure 2. CFPerm Methodology for detecting the distribution shift

Variants of Procedure

CFPerm0: There exists 1 feature whose original variable importance is larger than the maximal of the upper bound of the feature-level confidence level across each feature.

CFPerm1: There exists 1 feature whose original variable importance is higher than the 95% quantile(across features) with 99% upper level of the feature-specific confidence intervals.

Theorem1 Denote the permuted VIMPs $(v_{j,1}, ..., v_{j,B}), j \in \{1, ..., k\}$ and the **CFPerm0** procedure rejects H_0 when there $\exists j \ s.t. \ v_j > v_{i,l}, \ i \in 1, ..., k, \ l \in \{1, 2, ..., B\}$ under significance level α and number of permutations B. If $B > \frac{k}{\alpha} + 1$, then the type-I error control is attained.
Elaboration on the Assumptions: Under *concept drift*, the conditional outcome mechanism differs across batches, i.e., $P(Y_{new}|\mathbf{X}_{new}) \neq P(Y_{exist}|\mathbf{X}_{exist})$ so that $P(Y|X) \not\perp \mathbf{T}$. In this setting, the batch label captures systematic changes in the response surface and remains associated with Y even after conditioning on X, thereby violating the ignorability condition required for causal interpretation. Consequently, the proposed procedure should be viewed as a *distribution-shift detector* rather than a causal estimator: it detects departures from invariance in the data-generating process but does not identify a causal effect of T on Y.

Validity of the Proposed Procedure

- The theoretical guarantee is well-established for the conservative procedure **CFPerm0**. Type-I error is well-controlled(Figure 3 [a]-[b]).
- The test would achieve higher power in detecting concept drift(Figure 3[c]), covariate shift(Figure 3[d]) compared with the existing methods.
- The performance is highly competitive compared with the existing methods on a variety of **UCI** datasets (Table 1).

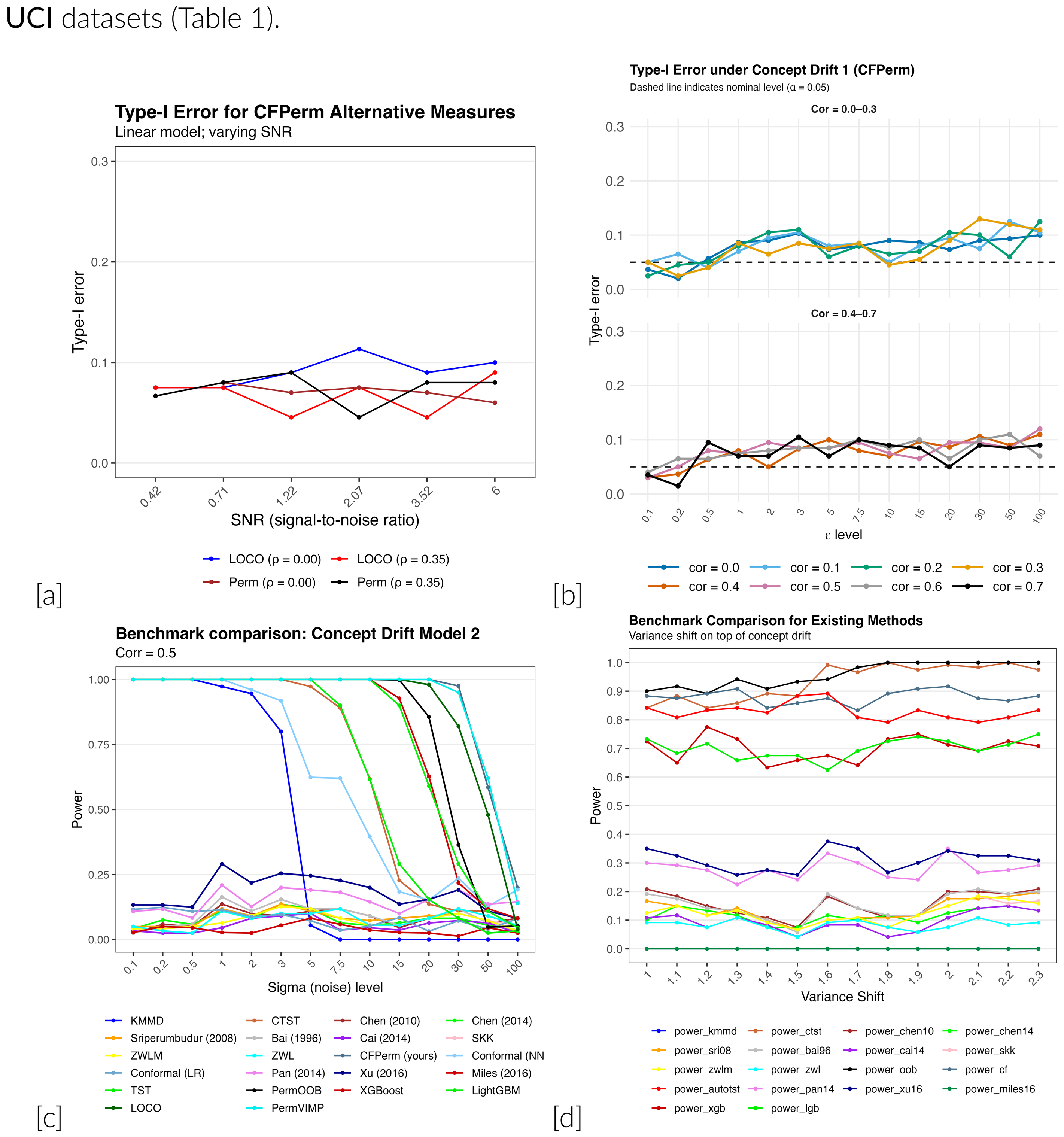

Figure 3. (a) Type-I error in models with different data quality (b) Type-I error w.r.t. different numbers of nuisance noise features (c) Power Comparison with existing methods for Concept Drift (d) Power Comparison with existing methods for Covariate Shift.

Data,Feature	CFPerm0	CFPerm1	CF_LOCO	CF_PermVIMP	autotst	c2st	chen14
Boston-no	0.310	0.910	0.860	1.000	1.000	0.945	1.000
Boston-cr	0.083	0.983	0.270	0.340	0.885	0.925	1.000
Boston-rm	0.235	0.835	0.780	0.820	0.835	1.000	0.685
Aq-V1	0.070	0.930	1.000	1.000	0.915	0.880	0.690
Au-V5	0.185	0.920	0.845	0.990	0.890	1.000	0.995
Au-V8	0.680	1.000	1.000	1.000	1.000	1.000	1.000

Table 1. Comparison of power on a variety of **UCI** datasets with different variable importance metrics.

Flexibility of the Framework - Attribution

The **CFPerm** framework is capable of identifying features that contribute significantly to the distribution shift. Suppose that H_0 is rejected; the features whose variable importance exceeds the decision threshold are selected as exhibiting significant distributional changes. As illustrated in Figure 4, $\mathbf{X}7$ and $\mathbf{X}8$ are identified as significantly shifted features.

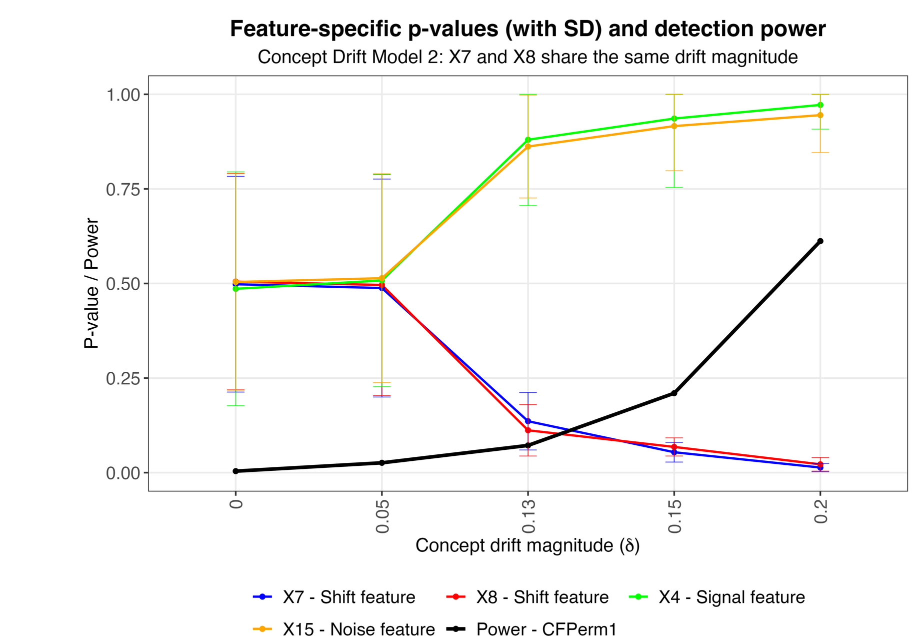

Figure 4. CFPerm for Selecting Features with Significant Contributions to Distributional Shift

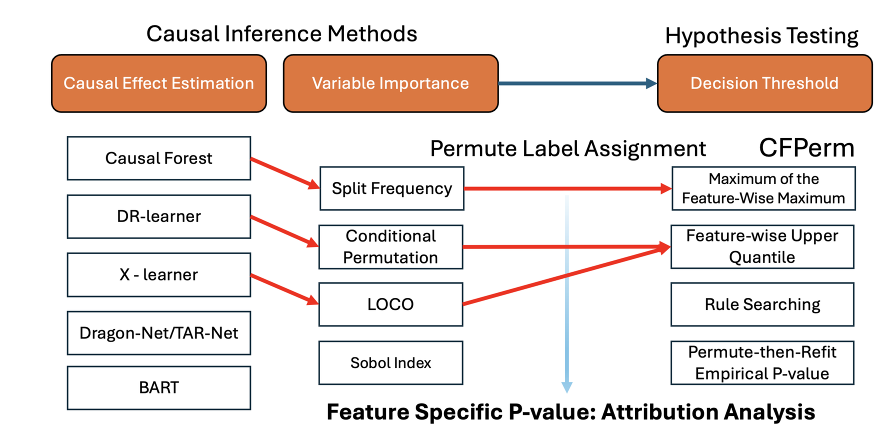

Figure 5. The Diagram for the proposed framework - Three-step Procedures

The proposed framework is highly flexible with the above three-step procedure(Figure 5).

Conclusion & Future Directions

- We propose a general-purposed framework with a class of hypothesis tests via leveraging causal inference methods to detect distribution shift with solid theoretical support. The **CFPerm** procedure of hypothesis testing is highly flexible, scalable and highly competitive in detecting distribution shift compared with existing methodologies.
- Feature importance estimates obtained from the fitted causal learner can be used as proxies for **variable importance** to characterize distributional shift.
- Extend the proposed **CFPerm** framework to the *granular characterization* of distribution shift via the **doubly robust pseudo-outcome learner**. Then followed by the *permute-then-refit* procedure to develop the test for distribution shift.
- **Disentanglement** of *Covariate Shift* and *Concept Drift* in Continuously Arriving Data Batches.

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